Interactive 3D Heritage Documentation System

for the Ifugao Bale House Scale Model

System Package · User Manual · Technical Documentation

Version 1.0 · Thesis Proof of Concept

1. Package Overview

This package contains all source files, documentation, and deployment instructions for the Interactive 3D Heritage Documentation System. The system is a browser-based 3D viewer built to document and present the Ifugao Bale house scale model created through photogrammetry as part of a thesis proof of concept.

1.1 What Is Included

File / Folder	Description
thesis-viewer.html	Main HTML page — open this to run the system
style.css	All visual styles and layout rules
viewer.js	All 3D viewer logic (Three.js, hotspots, tour, etc.)
images/	Folder for photographs of each Bale house component
models/	Folder for 3D model files (.ply or .glb)
README.txt	Quick-start instructions (also in this manual)
USER_MANUAL.doc	This document

1.2 System Requirements

Browser	Google Chrome 90+, Firefox 90+, Edge 90+, Safari 15+
Internet	Required for first load
Local server	Required only for local testing (see Section 3)
3D Model File	.ply or .glb format (stored in models/ or GitHub)
GitHub Account	Required for online deployment (free)
Git	Required only if pushing large files via Git LFS

NOTE *The system works entirely in a web browser — no Unity, Blender, or other 3D software is needed to run it. Blender was only used during the mesh preparation phase.*

2. Folder Structure

```
thesis-viewer/
← your main project folder
│
├── thesis-viewer.html
← open this in a browser to run
├── style.css
← visual styles
├── viewer.js
← all 3D viewer JavaScript
│
├── images/
← photos for hotspot panels
│   ├── roof-1.jpg
│   ├── roof-2.jpg
│   ├── walls-1.jpg
│   └── posts-1.jpg
│
├── models/
← 3D model files
│   └── Bale.ply
← (or Bale.glb)
│
├── README.txt
← quick-start guide
└── USER_MANUAL.docx
← this document
```

NOTE: *Keep all file and folder names exactly as shown. File names are case-sensitive on the web server (e.g., 'Bale.ply' and 'bale.ply' are different files).*

3. Opening the System Locally

Because the viewer loads a 3D model file and external scripts, you cannot simply double-click thesis-viewer.html in Windows Explorer — the browser will block the files for security reasons. You need to run a local web server.

3.1 Method A — VS Code Live Serve

This is the easiest method and requires no command line knowledge.

1. Download and install VS Code from code.visualstudio.com (free).
2. Open VS Code. Click Extensions (the puzzle icon on the left sidebar).
3. Search for 'Live Server' by Ritwick Dey and click Install.
4. In VS Code, click File → Open Folder and select your thesis-viewer/ folder.
5. In the file list on the left, right-click thesis-viewer.html.
6. Click 'Open with Live Server'.
7. A browser tab opens automatically at http://127.0.0.1:5500/thesis-viewer.html
8. The system is now running. Drag your .ply file onto the page to load the model.

3.2 Method B — Python

If Python is installed on your computer:

9. Open a terminal / command prompt.
10. Type: cd Desktop\thesis-viewer (adjust path to match your folder location).
11. Type one of these commands:

```
# Python 3 (most common):
python -m http.server 8080
# Python 2 (older computers):
python -m SimpleHTTPServer 8080
```

12. Open your browser and go to: http://localhost:8080/thesis-viewer.html

NOTE *To stop the server, go back to the terminal and press Ctrl + C.*

4. Open the system through the github pages

Open this link or copy paste to web browser: https://ljun00.github.io/thesis3d-viewer/thesis-viewer.html

5. How to Use the Viewer

This section explains every feature of the system from the end user's perspective

.5.1 Loading a Model

When the viewer first opens, you will see a dark screen with a drop zone. There are two ways to load a model:

– Drag and drop — drag your .ply or .glb file from your desktop and drop it onto the page.
– Browse — click the Browse file button and select the file from your computer.
– Auto-load — if MODEL_URL is set in viewer.js, the model loads automatically when the page opens.

NOTE *Supported formats: .ply (point cloud / mesh with vertex colours), .glb (textured 3D model from Blender), .gltf.*

5.2 Navigating the 3D Model

Control	Action
Left-click and drag	Rotate the model around its centre
Scroll wheel up / down	Zoom in or zoom out
Right-click and drag	Pan — move the view left, right, up, or down
Double-click	Zoom focus toward the clicked point
⌂ Reset View button	Returns the camera to the default position

5.3 Toolbar Buttons

Button	Function
▦ Wireframe	Shows the polygon mesh edges of the model. Click again to turn off.
⌂ Reset View	Returns the camera to its default starting position.
◑ Color	Switches between the original texture/vertex colours and plain grey.
☀ Brightness	Opens sliders for Exposure, Ambient light, and Sun light intensity.
↻ Auto Rotate	Slowly spins the model continuously. Click the model to stop.
⤢ Fullscreen	Expands the viewer to fill the entire screen. Press Esc to exit.
🧍 Scale Ref	Shows a human silhouette beside the model for size comparison.
▸ Tour	Starts the guided tour — camera flies to each part automatically.
+ Load File	Returns to the drop zone to load a different model file.

5.4 Hotspot Labels

When a model is loaded, three glowing green labels appear floating over the model:

- ▲ Atep · Roof — points to the thatched roof section
- ◈ Dingding · Walls — points to the wooden wall panels
- ● Tukkod · Posts — points to the foundation post section

Click any label to open the Part Information panel. The panel shows the component name, a description in English, a photo carousel, and cultural notes. Use the ‹ and › arrow buttons to browse multiple photos. Click ✕ to close the panel.

5.5 Guided Tour

27. Click the ▸ Tour button in the toolbar.
28. The camera flies automatically to the roof, pauses and shows the info panel, then moves to the walls, then the posts.
29. A counter (1 / 3, 2 / 3, 3 / 3) at the bottom tracks progress.
30. After all three stops, the camera returns to the default overview.
31. Click Stop Tour or drag the model at any time to exit the tour early.

5.6 Info Panel (Left Sidebar)

The left sidebar contains:

- About — brief project description
- How to Navigate — visual guide to mouse controls
- Toolbar Buttons — explanation of each button
- Model Information — live stats (format, vertices, faces, file size)

Click the ☰ button in the top-left header to show or hide the sidebar. On mobile, the sidebar collapses automatically and opens as an overlay.

8. Troubleshooting

Problem	Solution
Model not showing after drag & drop	Check browser console (F12 → Console). Make sure the file is .ply or .glb. Try a different browser (Chrome recommended).
Page is blank or shows only black	You may have opened the HTML file directly. Use Live Server or Python server (Section 3).
Model URL not loading	Make sure your GitHub repo is Public. Check that the URL starts with raw.githubusercontent.com or media.githubusercontent.com. Open the URL directly in a browser tab to test it.
Images not showing in hotspot panel	Check that photos are inside the images/ folder. File names must match exactly (case-sensitive). Open F12 Console — broken image paths appear as red errors.
Model appears underground	The model has been auto-placed on the grid. If it still looks wrong, this may be a scale or origin issue in the exported .ply/.glb file. Re-export from Blender with the origin at the base.
Hotspot labels in wrong position	Adjust the yFrac, xFrac, zFrac values in HOTSPOT_DEFS in viewer.js (Section 7).
Low FPS / laggy performance	Close other browser tabs. Reduce browser zoom to 100%. For very large models (>500k polygons), reduce detail in Blender before exporting.
Tour button not working	Make sure setupHotspots() was called and HOTSPOT_DEFS is defined in viewer.js. Check F12 Console for JavaScript errors.
GitHub Pages showing old version after update	GitHub Pages can take up to 5 minutes to update after a push. Do a hard refresh (Ctrl+Shift+R) to clear the browser cache.

9. Quick File Editing Reference

This table summarises the most common things you will want to change and exactly where to change them.

What to change	File and location
Auto-load model URL	viewer.js — line with: const MODEL_URL = ""
Part descriptions and names	viewer.js — PART_INFO object (roof, walls, posts)
Photos per part	viewer.js — images[] array inside each part in PART_INFO
Hotspot label text	viewer.js — HOTSPOT_DEFS — label: field
Hotspot positions	viewer.js — HOTSPOT_DEFS — yFrac, xFrac, zFrac values
Real height of house (scale)	viewer.js — MODEL_REAL_HEIGHT_M constant
Tour speed	viewer.js — TOUR_FLY_SEC and TOUR_DWELL_MS constants
Sidebar about text	thesis-viewer.html — <p class="about-text"> block
Page title in browser tab	thesis-viewer.html — <title> tag
Button colours and fonts	style.css — :root variables at the top
Toolbar button order	thesis-viewer.html — order of <button class="tbtn"> elements in #toolbar